Figure 3: Infrastructure, monitored services

Figure 4: Reports—All devices

To configure this host in Zenoss monitoring, under the *Infrastructure* menu, provide its IP address, and enter the required login name/password for the Windows PC. Configure the SNMP details, including the community (the default works for Windows). Select the services to be monitored. Once configured, the monitoring of the device and services begins. Out of the total of 111 available services to monitor, for test purposes we selected four: vsmon (ZoneAlarm), Avast (anti-virus), SNMP and Eventlog (see Figure 2 for an example). Go on adding as many devices, hosts and services that you want monitored. You can even use automatic network discovery, by giving a network subnet (e.g., 192.168.1.0/24 in our case) and add all the discovered IP addresses. (In this case, the network discovery was automatic, and the addition of two external firewall IPs was manual.) One of the internal Windows PCs was SNMP-enabled; all other devices were monitored only using ping.

## Reports

Once the set-up is complete and monitoring starts, the most important menu is *Reports*, to check the current status. Figure 4 shows the sample output for the All Devices report. In this, we observe that:

- 192.168.1.105, Ping and SNMP are both responding; for 192.168.1.211, there is a Ping response, but not for SNMP. That's because SNMP is not enabled on the CentOS machine.
- IPCop is configured to check only Ping response; SNMP is not configured properly on it since it's a test machine.
- The first firewall is down—there was no Ping response. (This is a live firewall on the Internet.) The second firewall is up, and SNMP is not configured on it.
- For *localhost*, *iptables* is configured to disable Ping responses; however, SNMP is responding.

Figure 5: Reports—Availability

As you can see in Figure 5, you can also check the availability reports for a range of dates that you specify.

Due to space constraints, I've covered only important menus. Do explore the various Zenoss menus on your own; you will find all your network monitoring requirements are catered for!

In a nutshell, Zenoss Core is a simple-to-install, easy-to-configure and rock-stable open source network-monitoring system. It can be deployed effectively to monitor various devices, hosts, and services. Zenoss and all its components are available under the GNU licence. Stack installers of Zenoss are available for all important Linux distros, including Red Hat, CentOS, Fedora, Ubuntu Server, OpenSuSE, SUSE Enterprise and Debian—as well as for several versions of Mac OS (available for free download). Its virtual appliances are also available for VMWare on Linux/Windows. END

### References


[1] Zenoss: *www.zenoss.org*
[2] Zenoss forum: *http://community.zenoss.org/community/forums*
[3] Zope application server: *www.zope.org*
[4] SNMP informant: *www.snmp-informant.com*
[5] Comparison of important network monitoring tools: [5] *http://en.wikipedia.org/wiki/Comparison_of_network_monitoring_systems*
Book: Zenoss Core Network and System Monitoring by Michael Badger



**By: Rajesh M Deodhar**

The author is a BE (Industrial Electronics), CISA (Certified Information Systems Auditor) and has a DCL (Diploma in Cyber Law). He has more than 15 years of experience in the field of computer hardware, networking, firewalls, and IS auditing as a director at Omega Systems and Services Pvt Ltd, Pune. Please send your feedback/queries to *rajesh@omegasystems.co.in*.